$$\begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 5 & 4 & 2 \\ 3 & 7 & 3 & 3 \end{bmatrix} \xrightarrow[+\text{Row }1]{\text{Row }2 = \text{Row }2} \begin{bmatrix} 1 & 2 & -1 & 4 \\ 3 & 7 & 3 & 6 \\ 3 & 7 & 3 & 3 \end{bmatrix} \xrightarrow[-\text{Row }2]{\text{Row }3 = \text{Row }3}$$

$$\begin{bmatrix} 1 & 2 & -1 & 4 \\ 3 & 7 & 3 & 6 \\ 0 & 0 & 0 & -3 \end{bmatrix} \xrightarrow[-3\times\text{Row }1]{\text{Row }3 = \text{Row }3} \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 1 & 6 & -6 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

Verification that the final matrix is in echelon form

1. There are no zero rows in the matrix. ∴ The condition that all the non-zero rows must be above the zero rows is vacuously True.

2. Row 1 Leading element $= 1$ in column 1
   Row 2 Leading element $= 1$ in Column 2
   Row 3 Leading element $= -3$ in Column 4

∴ The condition that for each row the leading element is in a column to the right of the ~~...~~ columns of leading elements of all rows above it is True.

3. Row 2 Column $1 =$ Row 3 Column $1 = 0$
   Row 3 Column $2 = 0$

∴ The condition that for each pivot column, all the elements below the pivot position is 0 is True.

∴ The matrix $\begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 1 & 6 & -6 \\ 0 & 0 & 0 & -3 \end{bmatrix}$ is in echelon form.

There exists a row in the matrix of the form $\begin{bmatrix} 0 & 0 & 0 & \dots & 0 & b \end{bmatrix}$ where $b \neq 0$. By Theorem 2, the corresponding system of equations is inconsistent, i.e. no solution set exists.

In Exercises 23 and 24, mark each statement True or False. Justify each answer.

23(a) Another notation for the vector $\begin{bmatrix} -4 \\ 3 \end{bmatrix}$ is $\begin{bmatrix} -4 & 3 \end{bmatrix}$

Solution: From Pg 24, Section: Vectors in R^2, a vector when represented in matrix form can only be represented as a